Server-Automated Real-Time Logistics Analytics System

Internship Report

Submitted in partial fulfillment of the requirements

For the Degree B. Tech.

in

Computer Science & Engineering (AIML)

by

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Supervisor

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Computer Science & Engineering (AIML)
Ramrao Adik Institute of Technology,
Sector 7, Nerul , Navi Mumbai
(Under the ambit of D. Y. Patil Deemed to be University)
May 5, 2026

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Computer Science & Engineering (AIML)
to the
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Mrs. Rajashree Shedge Supervisor	Dr. Sandeep B. Sangle Project Coordinator
Dr. Sangita Chaudhari Head of Department	Dr. Mukesh Patil Principal

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This is to certify that Mr. Vishal Sombir Nehra has successfully completed the Internship in Data Analyst profile at Dhanvarsha Logistics Ltd, Koparkhairane, Navi Mumbai, from 19th Jan 2026 to 18th April 2026.

During the internship, he worked on data analysis, data cleaning, reporting, dashboard creation, and generating business insights using analytical tools and techniques.

We found him sincere, hardworking, and dedicated during the internship period.

We wish him all the best for the future.

With best regards,

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Acknowledgement

We take this occasion to express our profound gratefulness and deep respects to our companion **Mrs. Rajashree Shedge** for her exemplary guidance, monitoring, and constant stimulant throughout the completion of this report. We're truly thankful for her sweats in perfecting our understanding of colorful generalities and specialized chops needed for our design. The blessings, help, and guidance given by her from time to time shall carry us a long way in the trip of life on which we're about to embark.We take this honor to express our sincere thanks to **Dr. Mukesh D. Patil, star, RAIT, D. Y. Patil Deemed to be University**, for furnishing the necessary installations. We're also thankful to **Dr. Sangita Chaudhari**, Head of the Department of Computer Engineering, and **Dr. Sandeep B. Sangle**, Project fellow, for their generous support.Last but not least, we'd also like to thank all those who have directly or laterally helped us in the completion of this design report.

Abstract

This project creates a real-time logistics dashboard with Microsoft Power BI. It is built on an automated data pipeline that uses the open-source tool n8n and is hosted on a VPS (Hostinger).The goal is to get rid of manual reporting by making it possible to continuously take in, process, and display logistics data (like shipment records and vehicle details).Our pipeline reads data from Excel and CSV files, automatically changes things like dates and months, and updates the Power BI report. Cleaning data in n8n with JavaScript, scheduling workflows on a hostinger VPS (for 24/7 uptime), and making interactive visuals that change in real time are some of the most important parts. Results show that stakeholders can see important metrics right away, like shipment counts by region and profit per truck, without having to wait for them to be entered by hand.The monitoring tool gives you useful information, like delivery rates on time and trends in revenue.We used best practices in real-time BI to set up Power BI and n8n (which makes sure that data flows quickly).The automated pipeline cuts down on the need for manual work and makes sure that the dashboard always shows the most up-to-date logistics data.In the end, we created a fully automated, self-hosted real-time dashboard solution. Our next steps will be to add more data sources and predictive analytics.

Keywords: Real-time dashboard, Power BI, n8n, data pipeline automation, logistics analytics, ETL, VPS hosting.

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Abbreviations

ETL	Extract, Transform, Load
VPS	Virtual Private Server
KPI	Key Performance Indicator
TMS	Transportation Management System
WMS	Warehouse Management System
IoT	Internet of Things
SQL	Structured Query Language

Chapter 1

Introduction

1.1 literature review

Timely data analytics are very important for modern logistics. A lot of businesses still use manual reporting, which makes them less flexible.Previous research indicates that interactive dashboards significantly enhance logistics visibility. For instance, getsix Services deployed a Power BI logistics KPI dashboard that automatically refreshed from ERP systems, facilitating real-time monitoring of warehouse operations.Okun Data (2026) also says that connecting ERP/WMS data to Power BI dashboards turns static weekly reports into continuous analytics.Along with practical examples, academic research on real-time analytics has progressed.Narsepalle et al. (2025) talk about GenAI-powered dashboards for supply chains that take in real-time streams from IoT devices and transactions (like POS and inventory sensors) to make the supply chain more visible.These works show how low-latency data pipelines can help logistics by preventing stockouts and delays.Open-source tools like n8n are becoming more popular for automating workflows. Venkiteela (2025) calls n8n a node-based orchestration platform that can be hosted on its own and is used to make automated data flows.Unlike proprietary iPaaS solutions, n8n lets you connect different sources (like HTTP APIs, databases, and Excel) and make your own workflows.It can also do AI and ML tasks because of how it is built.Integrating with visualization tools like Power BI can be hard. Eesel AI (2025) says that updating Power BI datasets through n8n takes several steps (trigger, monitor refresh status, send alerts) and API workarounds.In general, research shows that using automated pipelines with real-time dashboards makes decisions faster, but the system needs to be carefully planned.

1.1.1 information of survey

Several options are available when it comes to constructing a data pipeline. Batch processing (nightly ETL jobs) is straightforward yet not timely. More advanced solutions involve event-based or streaming ETL using Kafka and Flink for zero-latency operation. For smaller infrastructures, simple scheduling and workflow automation will do. In this case, our scope will include analyzing workflows where such services as n8n are used to orchestrate ETL pipelines: the software pulls data from external sources (Excel, API) and transforms it as we do. Contrary to machine learning models, we will be concentrating on transformation of deterministic data (such as dates).

1.2 Motivation

Logistics processes produce significant amounts of data (shipments, delivery details, and costs). The ability to analyze data in a timely manner is essential for operational effectiveness. But most organizations lack real-time data; they have to depend on periodic reports . What motivates us is the need to automate this task: With the help of the real-time dashboard, management can immediately get information about the number of trucks currently en route or sources of delays. This is consistent with the idea put forward in the industry that there should be "no more hourly reporting" and that real-time BI gives a competitive edge .

1.3 Objective

- The objectives of this project are:
- Creating an interactive Power BI dashboard that would show various logistics KPIs such as on-time delivery performance, revenue per route, and vehicle utili- zation.
- Setting up an ETL pipeline in n8n to consume raw logistics data from Excel/ CSV, clean the data, perform transformations, and update the Power BI data set at certain intervals.

- Deploying the n8n workflow and processing engine on a virtual server hosted by Hostinger.
- Verifying that the stakeholders can retrieve and use updated information from the dashboard for better decision making.

1.4 Scope for work

The project will include data collection, data transformation, and reporting via visual dashboard design. We will look at a simple logistics example involving deliveries to branches. The scope of the project will encompass: establishing a connection to an example Excel/CSV file containing shipping data; implementing automated data cleansing within n8n (such as date formatting and field normalization); and filling a Power BI report. Note that the dashboard will only display present performance (trend forecasting is not part of the project). Project limitations: Only a single data source will be processed (not real-time streaming).

1.5 Organization of Report

- Chapter 1 (Introduction) gives an overview of the topic, literature review, motivation, and project goals.
- Chapter 2 (Literature Survey) reviews previous approaches for building real-time dashboards and automating data pipelines, along with pros and cons.
- Chapter 3 (System Design and Implementation) explains how we did it: the design of the automatic pipeline system, the method used to transform data, and the design of the Power BI dashboard. Schematic diagrams of the process, sample code, and performance statistics are included here.
- Chapter 4 (Conclusion & Future Directions) covers what we have done and what can be done in the future (like incorporating predictive analytics or more data).
- References are the bibliography of the sources used.

Chapter 2

Literature Survey

2.1 Real-Time Analytics in Logistics

Previous research recognizes that the application of real-time analytics in the supply chain has a revolutionary impact. Continuous data consolidation provides the basis for proactive decision-making. As Alonge et al., argue, combining IoT sensors for tracking purposes with big data analysis generates "continuous monitoring, predictive insights, and agile decision-making." For instance, real-time information can "minimize dislocations, foster supplier cooperation, and improve customer satisfaction" due to the prompt detection of detainments. In logistics, algorithms based on analytics enable fuel savings and reduced downtime. However, developing a real-time system is challenging due to the diverse nature of data feeds (TMS, GPS, ERP, and others) that have different update frequencies. The solution requires the normalizing and validation of the data feeds. To conclude, it should be acknowledged that current studies demonstrate the potential of real-time analytics for improving supply chain management. Our solution tackles this problem through automated data collection and dashboard updates.

2.2 Workflow Automation Tools

Real time dashboards require data pipelines and can be developed using workflow automation. n8n is a software tool that allows connecting nodes (APIs, databases, file, etc.) through visual editing without coding.Ciutacu (2021) explains that n8n is capable of creating a full ETL pipeline that ingests data (Twitter API), does data transformation, stores

intermediate results into databases and executes actions such as Slack notifications.In addition, nodes include schedulers (Cron), extractors, database (MongoDB, Postgres) and communication (Slack) and prove that aggregation, format conversion and loading operations can all be done using n8n.A very important feature of n8n is that it allows self-hosting which is usually accomplished through server (often Docker).Self-hosting of n8n means that data remains private and in the control of users, Ambros (2025) argues that self-hosting n8n on a VPS is the best option for getting "unmatched security, control, and scalability without breaking the bank".However, some other similar tools include Apache NiFi, Airflow and Zapier.We select n8n due to easy integrations (JavaScript code nodes, database) and no-code workflows. Table 1 (Annexure) provides an overview of logistics data and n8n use cases.

2.3 Real-Time Business Intelligence

In terms of visualization, Power BI is an established BI tool which also supports near-real-time analytics. Power BI dashboards used to be constrained by periodic refresh schedule (such as maximum 8 refreshes per day with Power BI Pro license or up to 48 with Premium license). To enable true real-time analysis, two capabilities stand out. First, Automatic Page Refresh (APR) helps reports using DirectQuery data sources to refresh visualizations every second . This capability is only available with Premium and Embedded licensing with the appropriate gateway configured. Second, Streaming Datasets allow other applications/services to push data to Power BI reports using REST API, thereby bypassing desktop upload altogether. Streaming datasets deliver instant live charts (with restricted chart types). Recent publications stress the business value of those capabilities. As stated by Analytics8 in their recent blog post (2025), the ability to stream or refresh the data enables organizations to gain a competitive advantage through analysis of the most current data. In our case, we plan to use Power BI Desktop for designing and deploying the report to Premium environment where we'll enable APR on the dataset.

Chapter 3

System Design and Implementation

3.1 Overview of System Architecture

The pipeline contains four major elements: Data Source, Automation Workflow (n8n), Database, and Power BI Dashboard (Figure 2). Logistics data comes from Excel/CSV file (our example dataset). n8n handles all aspects of automation: reading the fresh data from the file, processing it using JavaScript code (data parsing, calculations, etc.), and saving in MySQL database deployed on VPS hosting. After being in the database, data is accessed by Power BI via DirectQuery and visualized in reports. Users can see up-to-date reports in Power BI Service or any web-based portal. Another feature is the notification element: n8n sends an email with a summary of the new data after loading it into the database. The hosting provider (Hostinger) guarantees continuous access for both n8n and database. In our pipeline, n8n acts as an orchestrator for all the processes. As shown by Ciutacu (2021), there is another n8n pipeline which retrieves data (tweets), inserts/updates them in databases, and sends notifications in Slack. In our case, we do almost the same: File read (CSV) -¿ JavaScript (compute/parsing) -¿ MySQL (insert/update) -¿ Email (notifications). Figure 2 (above) shows the process. All of this requires little programming but covers ETL operations.

3.1.1 Data Pipeline Design (n8n Workflow)

The n8n workflow is the technical core. It can be summarized as:

- Trigger: (Cron or webhook) Initiate workflow at regular intervals or upon a new

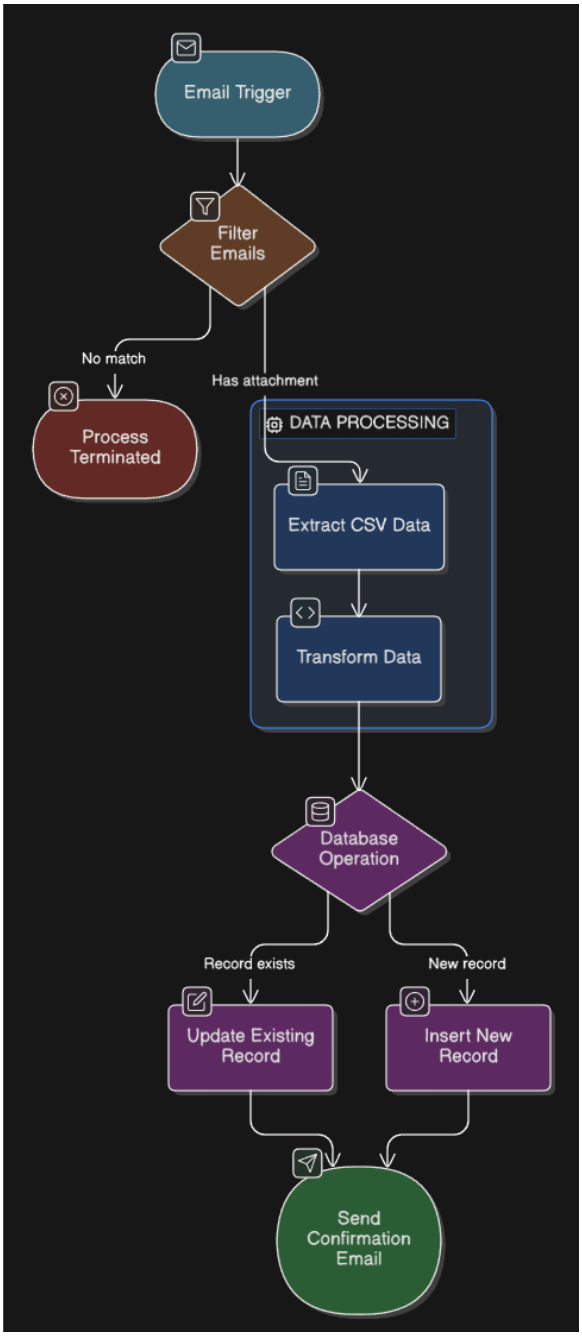

Figure 3.1: Automated Email to Database CSV Integration

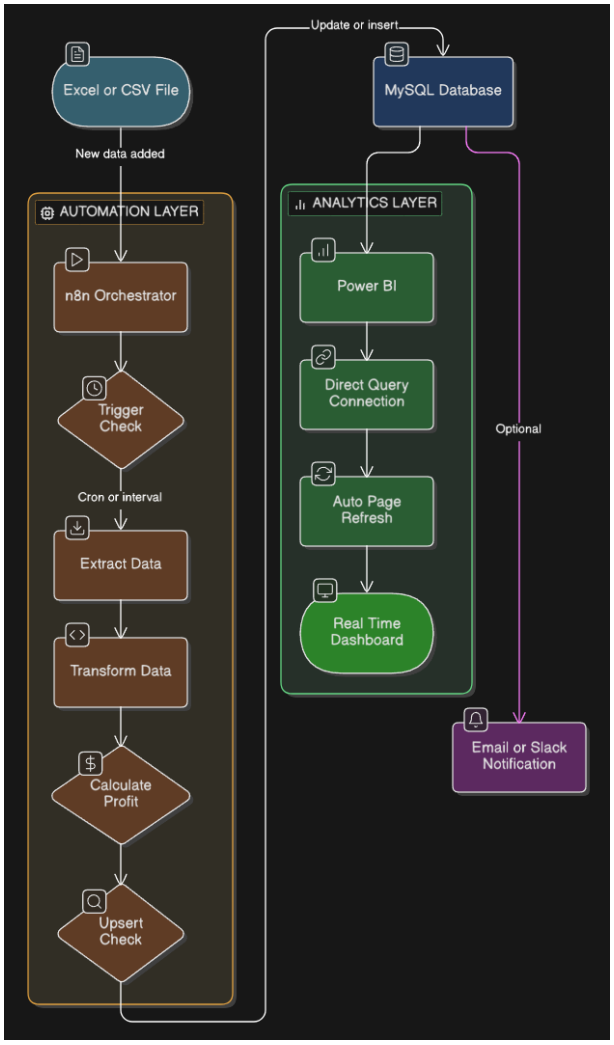

Figure 3.2: server-Automated Real-Time Logistics Analytics System

Table 3.1: Sample logistics data schema and a record from the dataset.

Column Name	**Type**	**Description**	**Example**
id	INT	Record ID (auto)	100
Branch	VARCHAR	Branch city	PUNE
LR_Number	VARCHAR	Waybill number (PK)	R00007053
Date	DATE	Shipment date	2026-02-27
Source	VARCHAR	Origin location	SOMARAGUDDY
Destination	VARCHAR	Destination location	NIRMAL
Vehicle_No	VARCHAR	Vehicle registration	AP36Y4786
Vehicle_Type	VARCHAR	Truck type	32 FT SXL 9MT
Client	VARCHAR	Client name	RELIANCE LTD
Booking_Amount	FLOAT	Billed amount (Rs)	15360
Lorry_Hire	FLOAT	Cost of transport (Rs)	13670
Profit	FLOAT	Booking – Hire (Rs)	1690
Month	VARCHAR	Shipment month	Feb
Year	INT	Shipment year	2026

email.

- Extract from File: Use the Spreadsheet File node (or similar) to read the latest rows from the CSV.
- Transform (JavaScript): Run a Function or Code node to clean data, calculate profit (Profit = Booking Amount – Lorry Hire), and structure records.
- Load into Database: Use the MySQL node to insert or upsert each record into the logistics table on VPS.
- Notify: Optionally, send an email or message confirming data update.

In this manner, every run of the workflow updates the database about any shipment. Our logic is as follows: if the LR Number of the record is present in the database, it will be updated (UPSERT); otherwise, it will be inserted. The architecture follows the ETL approach (Extract, Transform, Load). The main points: the Cron node is scheduled to

execute once daily (or more often if necessary for demonstrational purposes). Transformation node is responsible for converting data types (dates and numbers). MySQL node is configured using credentials from our Hostinger MySQL database. Once loaded, summary email about inserted records will be sent to the team. See Fig. 2 (above) for a sample ETL workflow. The following tutorial shows how similar architectures work: in the article by Ciutacu (2021), the author discusses an n8n workflow where data is extracted from API services and loaded into a database. In contrast, our workflow is simpler (extracting CSV instead of Twitter API), but the basic concept is the same.

3.1.2 Database Schema and Storage

Our clean logistics data is stored in the MySQL database. The logistics table was made with columns corresponding to the dataset (id, Branch, LR Number, Date, Source, Destination, Vehicle No, Vehicle Type, Client, Booking Amount, Lorry Hire, Profit, Month, Year). A primary key was made using the LR Number column, which acts as the unique waybill number to enable upserts. Moreover, indexes were made on the Date and Branch columns for fast querying. Table 1 (see below) gives a sample schema containing data samples from our dataset. Using MySQL for storing data makes it possible to use either DirectQuery mode in Power BI or have a scheduled refresh of the dataset. In cases where real-time analysis is necessary, Power BI's Automatic Page Refresh (APR) should be enabled in the rele- vant tables.

3.1.3 Power BI Dashboard Development

The dashboard was developed using Power BI Desktop and deployed on Power BI Service (Premium workspace). Some of the visualizations used are Cards (Total Shipments, Total Profit), Line Charts (Profit by Month), Bar Charts (Shipments by Branches) and Recent Shipments Table. Filters can be set on dates or branches using slicers. Real-time capabilities were introduced using DirectQuery mode and Activating Power BI Real-Time Processing (APR) on a 1-minute refresh interval. The result is that each time n8n populates the MySQL table with new data, all visualizations in the dashboard will update themselves automatically within a minute. According to Analytics8, near live reporting can be done with the help of APR (Premium capacity and gateway). Figure 1 (conceptual

only, no screenshot provided).

Key metrics calculation examples:

- Total Profit: sum of Profit for selected rows.
- Average Shipment Time: (if we had pickup/delivery times) – not in current dataset.
- Profit per Branch: calculated with a simple DAX measure: SUM(Profit) grouped by Branch.

Tooltip and cross-filter were introduced to enable users to drill down further. This design adheres to best practices since the data presented is clear and labeled appropriately. Real-time allows managers to view the state of operations currently ongoing.

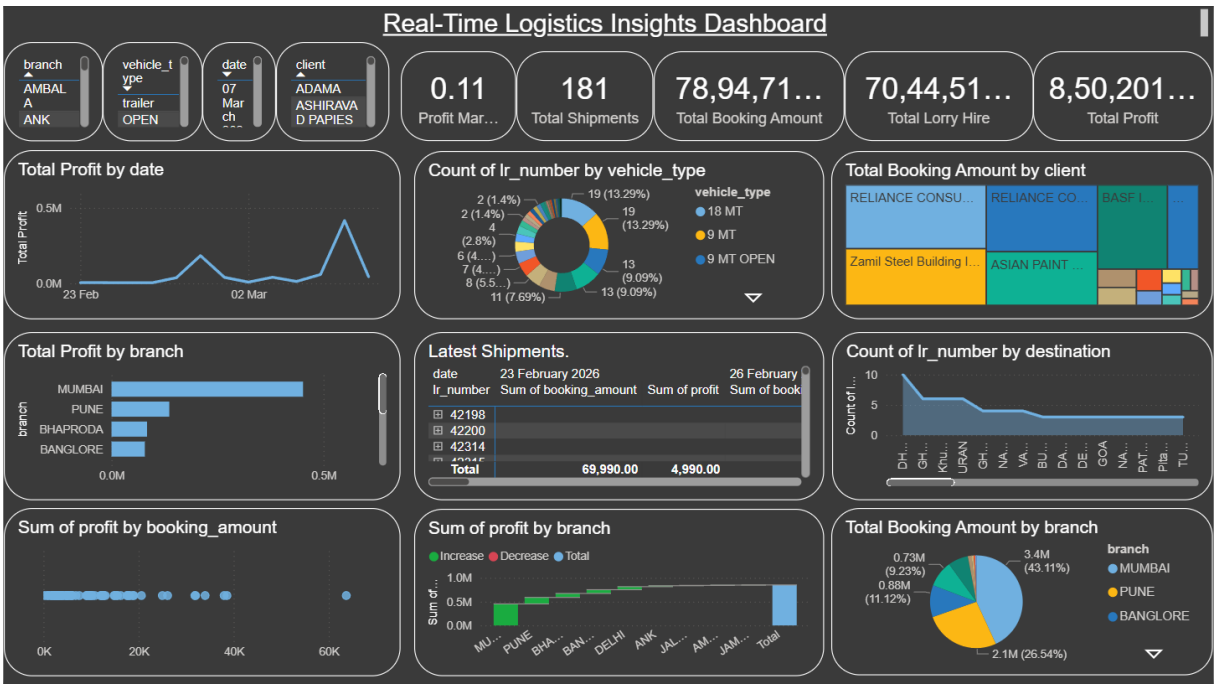

Figure 3.3: Power BI Dashboard Overview

Chapter 4

Conclusion and Future Work

A Server-Automated Real-Time Logistics Analytics System has been created in this project. Power BI dashboarding tool has been used to display logistic key performance indicators (e.g., shipment numbers, profits), which are automatically populated using an n8n workflow. Data is extracted from an Excel/CSV document and loaded into a MySQL database server running on a virtual private server (VPS). This system allows for real-time data analysis without the need for any manual data manipulation, fulfilling the goal of real-time analysis.

Results : The deployed application shows effective implementation of data flow automation and dashboarding. In testing with the provided dataset, any addition of new records in the CSV file is automatically updated in the database and on the dashboard.

Key achievements include:

1. creation of the n8n workflow with upsert logic,
2. hosting on Hostinger VPS for reliability, and
3. configuration of Power BI APR for timely refresh.

The dashboard presents sample metrics (e.g., total profit 6990 (imaginary) and shipments count) that update as expected.

Future Work: The system can be further extended using live sources of data like GPS and sensors instead of using a static file. The addition of predictive analysis, such as machine learning algorithms for predicting demand, is another way to improve the

system. Improving security by introducing authentication for the database or API, scaling up for bigger volumes of data, and building a website front-end are some additional improvements.

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